

Air France Flight 296

Flight model: Airbus A-320
Date: 26 June,1988

From: Alsace, France
To:Alsace, France

The Airbus A320 was a new jet back in the period. The flight used for an air show demonstration. The passengers on the plane was around 130. The Airbus A320 was the first flight to use fly by wire. Airbus was first to introduce it in commercial planes.

Problem

During Day, the pilots tries to perform a maneuver called "Low altitue nose up alpha fly by max". This air field was too small and thus was not mapped in the flight computer. By the time the pilots found the air field they were too late and had to descend steeply to reach the run way and must loss speed to perform the maneuver. Then the pilot reaches the altitude and disables the auto throttle so that the plane would not increase the speed of the plane when performing the maneuver. Right before taking off again, the pilots find a forest in front. The pilots tries to apply full power to the engines but the plane crashes into the forest. The fuselage was intact and thus everyone had survived.

Investigation

Right before take off after the maneuver when the pilots tries to increase the power to full, the footage captured by an reports shows that the plane has not taken off or achieved the required height to be safe after the maneuver. It was also found the minimum altitude set by Air France for the maneuver was too low when compared to regulation. Also, the maps were prepared and handed over to the pilots 48 hours before the event which again lacked the forest area information. But it was found that the average height of the forest was 40 feet but the plane was asked to flight above 100 feet. The investigators find that the flight has no mechanical failures and had diverted from the flight plane that the pilot had planed. It was found that instead of maintaining a stable altitude the plane started to loss altitude when performing the maneuver. The pilots right before the maneuver found that the crowd were standing on different runway than the earlier planned one therefore the pilots had to change their plans immediately to perform the maneuver. This distracted the pilots who then forgot to monitor the plane altitude. The plane also had another altimeter which showed the value of the altitude on an digital display which the pilots had claimed it was difficult to read. Apart from that the plane also had a digital voice call-outs of the altimeter which the pilots claim they were not able to hear the same. When asked about what the pilots did when they saw the tree, they replied that they tried to increase the power, but the plane failed to respond. The pilots also cite from a document by the manufacturer that the plane performance may degrade at low altitude due to lower air flow thus the engine cannot accelerate. The pilots say that when he gave the input to pull the aircraft put using the control column, the plane computer had overwritten his input and was pushing down. His accusations were the Airbus had given more control to the computer. His accusations were supported using the data from the flight data recorder where in when the planes side stick were pulled up the computer instructed elevators to decrease the altitude. The **side sticks** are the ones that control the elevators. The investigators say that the altitude and speed together set the plane automatically to landing mode. When perform a simulator test it has confirmed that the plane had triggered a nose down response to the pilots input. But the plane had not gone into landing mode rather entered into a main feature **stall protection**.

Stall protection bring the nose down to prevent the plane from stalling, which happens when the engine losses airflow. But the investigators say this prevented the plane from crash before the runway ends.

Changes that were brought after the incident

The investigators report suggested ban on passengers in all demonstration flights.

Reference

1. Pilot VS Plane — Faulty Readings — FULL EPISODE — Mayday: Air Disaster
<https://www.youtube.com/watch?v=jxdHUU2nUYk>